Quantitative Research Methods III

CASE: "AN EMPIRICAL STUDY OF EDUCATIONAL OUTCOMES"
Regression Evidence from PISA 2022 Across Eight Countries

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Question 0 – Data preparation & Standardization

The standardization leads to unitless variables with a mean of 0. This allows for different variables to be compared in terms of standard deviations from the mean. As seen in *figure 1*.

Question 1 - Univariate Data Visualization

Table 1 shows the distribution of the standardized test scores for mathematics, reading and science. The modus is negative for all distributions. Most observations lie between approximately -2 and +2 standard deviations from the mean. Virtually no observations are more than 3,75 standard deviations removed from the mean. All distributions are right skewed, because the number of counts that exceed the +2,5 standard deviation is greater than the number of counts that exceed the -2,5 standard deviations. The math test score graph is most right skewed while the reading test score graph is least right skewed. We conclude that the distributions are not perfectly normal. There are more students with extreme high scores then students with extreme low scores.

Question 2 - Bivariate Data Visualization

Figure 2 shows a strong positive correlation between the standardized reading scores and the standardized math scores with a correlation coefficient of just below 90%.

Question 3 – Univariate Regression

We used a standard error and a robust standard error to determine the significance of the coefficient. From question 4 onwards, we only use robust standard errors.

H_0: $\boldsymbol{\beta_1 = 0}$ & H_1: $\boldsymbol{\beta_1 \neq 0}$

To analyze these hypotheses we used the t-test. The resulting *table 2* shows that the p-value is much lower than the usual $\boldsymbol{\alpha = 0,05}$. We therefore conclude that the coefficient ($\boldsymbol{\widehat{\beta}_1}$) of 0,895 is statistically significant and we reject the null hypothesis. This value can be interpreted as follows: when a student's standardized reading score increases by 1, their expected standardized math score increases by an average of 0,895.

H_0: $\boldsymbol{\beta_0 = 0}$ & H_1: $\boldsymbol{\beta_0 \neq 0}$

To analyze these hypotheses we used the t-test. The resulting table shows that the p-value is much higher than the usual $\boldsymbol{\alpha = 0,05}$. We therefore conclude that the coefficient ($\boldsymbol{\widehat{\beta}_1}$) of 0,000 is statistically insignificant and therefore we cannot reject the null hypothesis. This means that the regression passes through the origin. From the regression in *table 2*, we cannot conclude causal relationship. We suspect that there to be an omitted variable such as general intelligence that correlates with both test scores.

Question 4 - Multivariate Regression

We use the following dummies for the regression in *table 3*:

D_1=1: mostly speaks the test language

D_1=0: mostly does not speak the test language (this is the reference group).

In *table 3* we observe that both slope coefficients ($\hat{\beta}_1$ and $\hat{\beta}_2$) and the constant ($\hat{\beta}_0$) are statistically significant. For the students do not mostly speak the test language we

find a constant of 0,031. For the students that do mostly speak the test language we find a combined constant of 0,031 – 0,038 = -0,007; implying that people that speak the language at home score 0,038 less than those that do not, keeping the reading score constant. In this model, both groups have the same slope coefficient of the standardized reading score of 0,898, meaning that if their reading score increases by 1, their expected standardized math score increases by 0,898.

This result is somewhat puzzling, since you would expect people that mostly speak the test language to perform better on the math tests given that they are more likely to understand the questions better. Maybe this has something to do with the fact that math scores are less likely to be influenced by language abilities compared to reading scores.

Question 5 – Interaction

Here we have added the interaction term of Reading × Speaks Test Language to the model shown in *table 4*. All slope coefficients ($\hat{\beta}_1$, $\hat{\beta}_2$ and $\hat{\beta}_3$) and the constant ($\hat{\beta}_0$) are statistically significant. The reference group are the students that don't speak the test language at home. For the reference group we find a negative constant of -0,034 ($\hat{\beta}_0$) and a coefficient of standardized reading score of 0,759 ($\hat{\beta}_1$,). This means that when their standardized reading score increases by 1, their expected standardized math score increases by 0,759 = the slope coefficient.

For the non-reference group (speaks test language at home) we find a coefficient of the dummy variable Speaks Test Language at Home of 0.026 and for the coefficient of interaction term we find a coefficient of 0,155 ($\hat{\beta}_3$). This leads to the conclusion that the intercept of the non-reference group is -0,034 + 0,026= -0,008, and the slope coefficient of 0,155+0,759 = 0,914. The slope of the regression for the non-reference group becomes steeper and shifts upwards in totality.

These result corresponds to our initial expectations. The relationship between the reading scores and math scores seem to be stronger for the people that speak the test language at home.

Question 6 – Interaction

In *table 6* we notice the intercept and all dummy coefficients are statistically significant with extremely low p-values. The reference group in this model is the Dominican Republic. Keeping the country-of-origin constant, an increase of 1 in ICTRES, increases the average standardized math score by -0,240.

Question 7

In *table 7* we observe that coefficient of ICTRES is 0,0545 and statistically significant which means that given that ICTRES increases by 1, that the average math score is multiplied by $e^{0,0545} \approx 1,0545$. The dummy coefficient of 0,273 for Türkiye implies that the average math score of students from Türkiye is multiplied by $e^{0,273} \approx 1,314$. Log-transformed models are more useful when the relationship you are analyzing is exponential. The logged model might give a better fit in the regression if there is a non-

linear relationship between ICTRES and math scores. However, if this relationship is linear the resulting regression will be worse and the approach from question 6 will be better.

Question 8

In *table 8* we observe that the more food people have available to them the higher the dummy variable coefficient is increasing the corresponding groups' intercept, meaning that the richer somebody is, the higher you expect their math score to be. However, the coefficient of ICTRES has hardly changed (0,2404 → 0,2375).
This difference is the omitted variable bias = 0,0029. In the according generated output, poverty has a negative influence on math scores. Since the omitted variable bias is positive we can conclude form the formula:
$\beta_2 \frac{n \cdot Cov(X_1, X_2)}{\sum(X_{1i} - \bar{X}_1)^2}$; that the correlation between ICTRES and poverty is negative.

Question 9

The relevance condition states Corr(Z, X) ≠ 0, meaning there should be a non-zero correlation between HISCED and ICTRES. We expect there to be a correlation between education level and ICT resources due to the fact that higher education is likely to lead to higher income and therefore more ICT facilities.
The exogeneity condition states corr(Z,u) = 0, meaning there should be no correlation between the error term and HISCED. We don't expect this to hold since intelligence of the student could be an omitted variable which is correlated with both the level of education of the parents and the ICT facilities at home (when taking the previous correlation explanation into account).

Question 10

Given that all the control variables are kept constant, we would expect the average standardized math score to increase by 0,418 if ICTRES increases by 1. *Table 9* however, shows us the exogeneity condition is violated, this relation is more likely to be associational rather than causal. Given the formula of the omitted variable bias:
$\beta_2 \frac{n \cdot Cov(X_1, X_2)}{\sum(X_{1i} - \bar{X}_1)^2}$; we would expect there to be a positive correlation between intelligence of the student (=omitted variable) and the educational level of the parent. Additionally, we expect there to be an positive correlation between the educational level of the parent and the ICT facilities. Therefore, ICT facilities are likely to be positively correlated with an omitted variable creating a positive covariance. Furthermore, we expect intelligence to positively affect the standardized math scores so β_2 is likely to be positive. Therefore, the omitted variable bias is positive as well.

Appendix:

Table 1: Summary Statistics of Original and Standardized Test Scores

Score	Mean_original	SD_original	Mean_standardized	SD_standardized
Math	416.40	90.51	0	1
Reading	418.12	94.29	0	1
Science	430.70	92.87	0	1

Note: Standardized scores are calculated using Z = (X − mean(X)) / sd(X). After standardization the mean equals 0 and the standard deviation equals 1.

Figure 1:

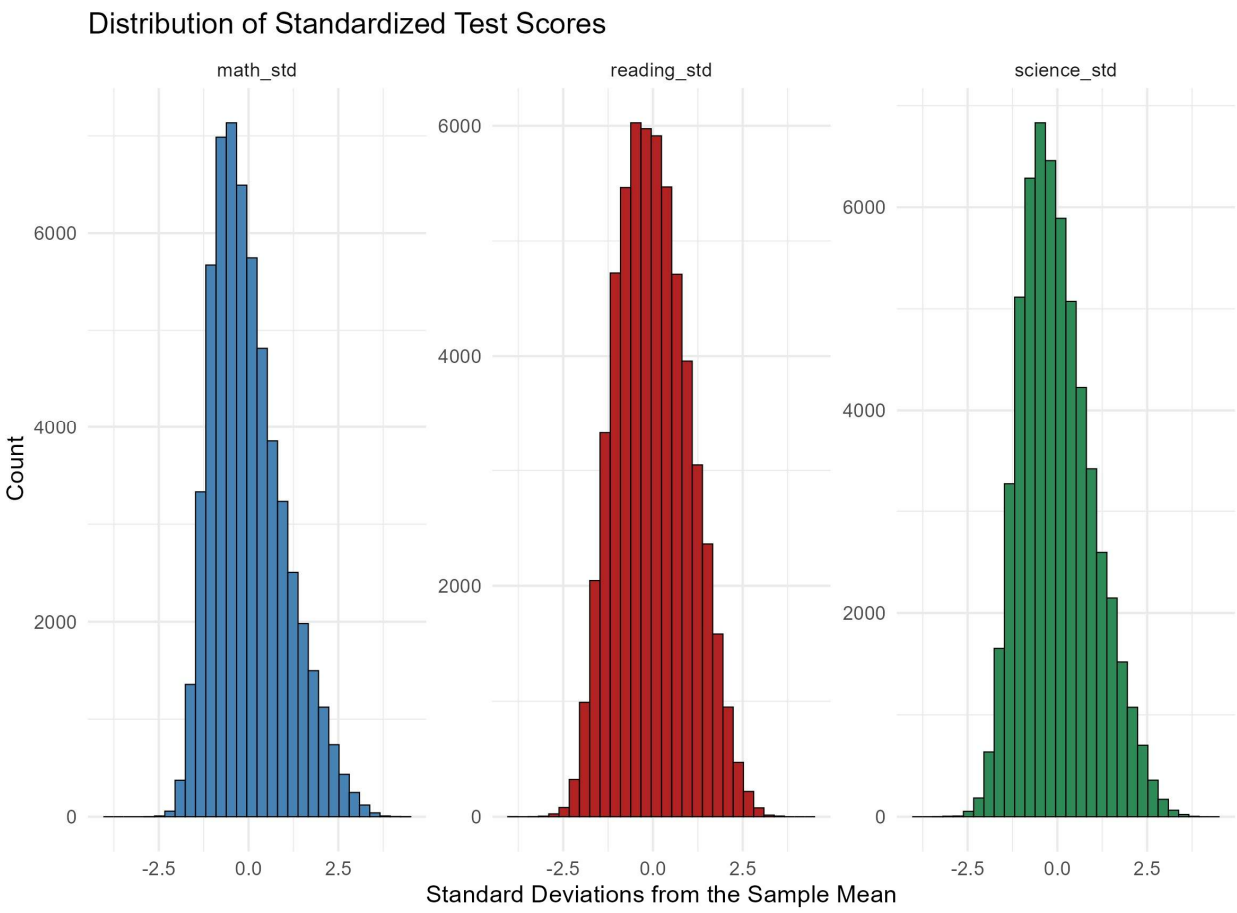


Figure 1.
Note: The figure shows the distribution of standardized mathematics, reading, and science scores. Standardization is performed using z-scores, resulting in a mean of zero and a standard deviation of one.

Figure 2:

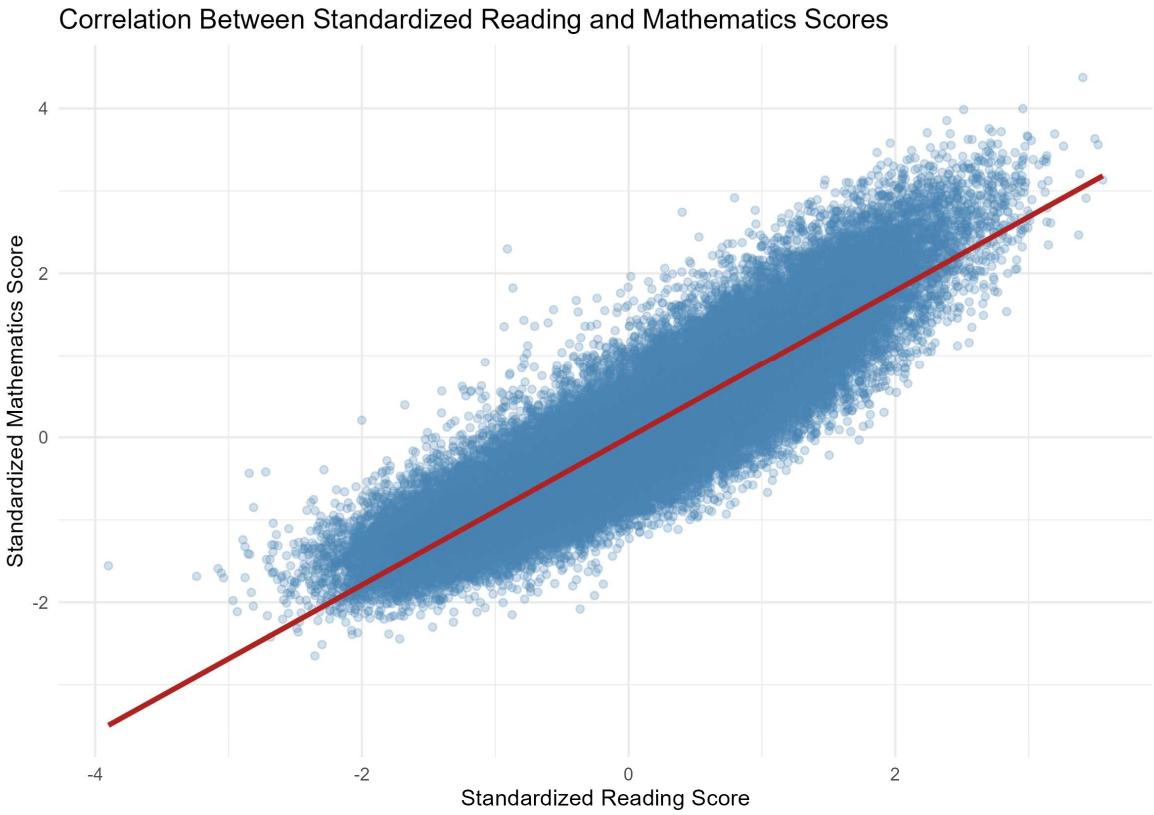


Figure 2.
Note: Each point represents a student in the sample. The solid line shows the fitted linear relationship between standardized reading and mathematics scores.Both variables are expressed as z-scores, meaning that values represent deviations from the sample mean measured in standard deviations.

Table 2. Regression Results for Standardized Mathematics Scores

	Conventional SE	Robust SE (HC1)
Constant	0.000	0.000
	(0.002)	(0.002)
Standardized Reading Score	0.895***	0.895***
	(0.002)	(0.002)
Observations	57768	57768
R-squared	0.801	0.801
Adjusted R-squared	0.801	0.801

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Model (1) reports conventional OLS standard errors.

Model (2) reports heteroskedasticity-robust standard errors (HC1).

The dependent variable is the standardized mathematics score.

Table 3: Multivariate Regression Results for Standardized Mathematics Scores

	(1)
Constant	0.031***
	(0.004)
Standardized Reading Score	0.898***
	(0.002)
Speaks Test Language at Home	-0.038***
	(0.005)
Observations	57104
R-squared	0.801
Adjusted R-squared	0.801

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

The dependent variable is the standardized mathematics score.

The variable 'Speaks Test Language at Home' equals 1 if the student speaks the test language at home and 0 otherwise.

Table 4: Interaction Regression Results for Standardized Mathematics Scores

	(1)
Constant	-0.034***
	(0.005)
Standardized Reading Score	0.759***
	(0.006)
Speaks Test Language at Home	0.026***
	(0.005)
Reading × Speaks Test Language	0.155***
	(0.006)
Observations	57104
R-squared	0.803
Adjusted R-squared	0.803

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

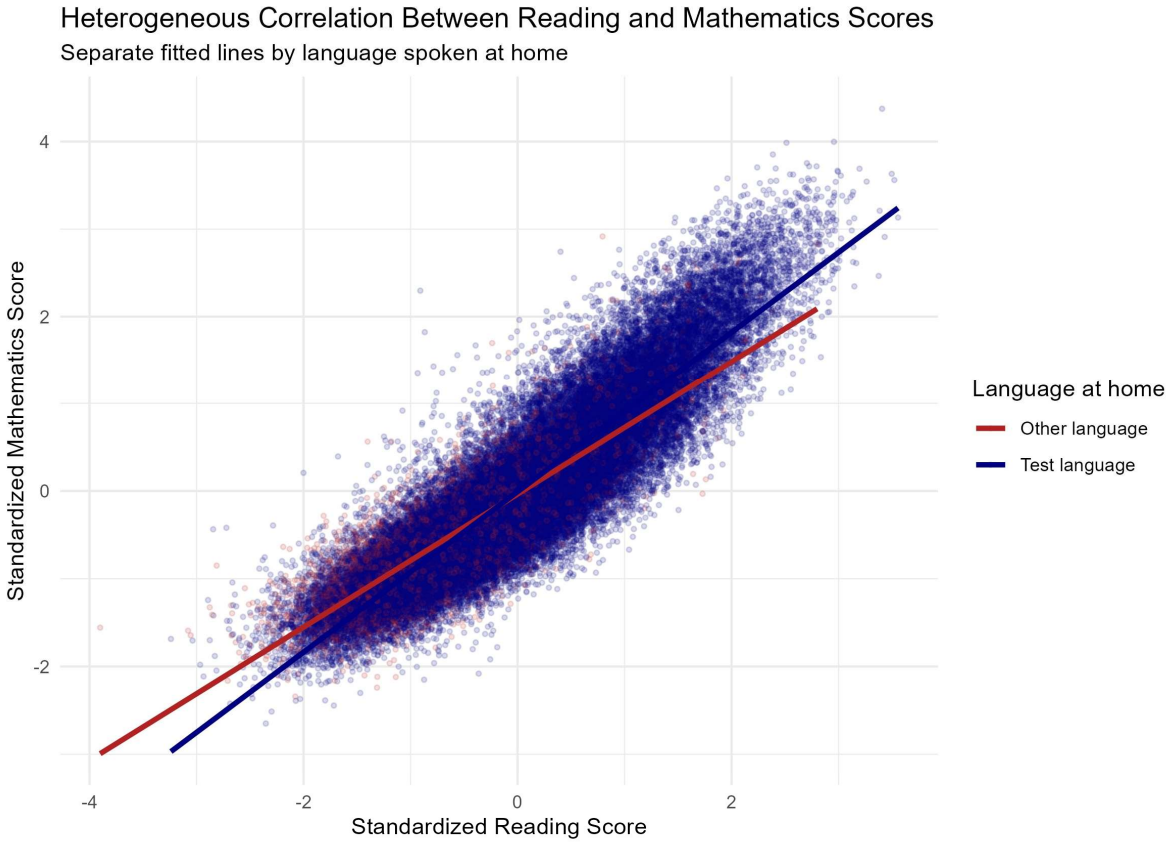
The dependent variable is the standardized mathematics score.

The variable 'Speaks Test Language at Home' equals 1 if the student speaks the test language at home and 0 otherwise.

The interaction term captures whether the relationship between reading and mathematics differs by language spoken at home.

- β_1 = effect of reading_std, when speaks_test_lang =0
- β_3 = difference in the effect between the two groups
- $\beta_1 + \beta_3$ = effect when, speaks_test_lang = 1

Figure 3:



Note: Each point represents a student. The fitted lines show the estimated linear relationship between standardized reading and mathematics scores for students who speak the test language at home and those who do not.

Table 5: Regression Results for Standardized Math Scores Part 1

Table 5. Regression Results for Standardized Mathematics Scores

	Dependent variable:		
	Standardized Mathematics Score		
	Q3	Q4	Q5
	(1)	(2)	(3)
Standardized Reading Score	0.895***	0.898***	0.759***
	(0.002)	(0.002)	(0.006)
Speaks Test Language at Home		-0.038***	0.026***
		(0.005)	(0.005)
Reading × Speaks Test Language			0.155***
			(0.006)
Constant	0.000	0.031***	-0.034***
	(0.002)	(0.004)	(0.005)
Observations	57,768	57,104	57,104
R^2	0.801	0.801	0.803
Adjusted R^2	0.801	0.801	0.803

Note: *p<0.1; **p<0.05; ***p<0.01

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses. The variable 'Speaks Test Language at Home' equals 1 if the student speaks the test language at home and 0 otherwise.

Table 6:Regression Results for Standardized Mathematics Scores and ICT Resources

	(1)
Constant	-0.549***
	(0.008)
ICT Resources at Home	0.240***
	(0.004)
Observations	45306
R-squared	0.268
Adjusted R-squared	0.268

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

The dependent variable is the standardized mathematics score.

Country dummies are included in the regression but omitted from the table.

Table 7: Regression Results for Log Mathematics Scores and ICT Resources

	(1)
Constant	5.886***
	(0.002)
ICT Resources at Home	0.054***
	(0.001)
Observations	45306
R-squared	0.267
Adjusted R-squared	0.267

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

The dependent variable is the log-transformed mathematics score.

Country dummies are included in the regression but omitted from the table.

Table 8: Regression Results with Food Availability Control

	(1)
Constant	-0.746***
	(0.021)
ICT Resources at Home	0.238***
	(0.004)
Observations	39210
R-squared	0.322
Adjusted R-squared	0.321

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

The dependent variable is the standardized mathematics score.

Country dummies and food availability categories are included in the regression.

Country dummy coefficients are omitted from the table.

Table 9: Instrumental Variables Regression Results for Mathematics Scores

	(1)
Constant	-0.454***
	(0.028)
ICT Resources at Home	0.418***
	(0.011)
Observations	39105
R-squared	0.277
Adjusted R-squared	0.277

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

The dependent variable is the standardized mathematics score.

ICT Resources at Home is instrumented using parental education.

Country dummies, food availability categories, and parental education dummies are included in the estimation but omitted from the #table.

Table Part 2: Regression Results

	Q6	Q7	Q8	Q10
Constant	-0.549***	5.886***	-0.746***	-0.454***
	(0.008)	(0.002)	(0.021)	(0.028)
ICT Resources at Home	0.240***	0.054***	0.238***	0.418***
	(0.004)	(0.001)	(0.004)	(0.011)
food_shortage4 to 5 times a week			-0.198***	-0.203***
			(0.038)	(0.042)
food_shortageAbout once a week			0.054*	0.053*
			(0.023)	(0.025)
food_shortageEvery day or almost every day			-0.185***	-0.196***
			(0.025)	(0.027)
food_shortageNever or almost never			0.248***	0.151***
			(0.020)	(0.022)
Observations	45306	45306	39210	39105
R-squared	0.268	0.267	0.322	0.277
Adjusted R-squared	0.268	0.267	0.321	0.277

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Notes:

Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.

The dependent variable is the standardized mathematics score in Q6, Q8, and Q10.

In Q7 the dependent variable is the log-transformed mathematics score.

Country dummies are included in all specifications.

Food availability and parental education controls are included where relevant.

R Code

```
#rm(list = ls())
#graphics.off()
#cat("\014")

## REMEMBER to install load packages explicitly for pdf knitting later!!
#install.packages(c("readr","dplyr","ggplot2","sandwich","lmtest","AER","modelsummar
y","forcats","tidyr","scales", "lubridate","moments","kableExtra","tinytex", "ggtext",
"stargazer", "flextabl", "officer", "stringr", "pandoc"))

library(readr)
library(dplyr)
library(ggplot2)
library(sandwich)
library(lmtest)
library(AER)
library(modelsummary)
library(forcats)
library(tidyr)
library(scales)
library(lubridate)
library(moments)
library(kableExtra)
library(tinytex)
library(ggtext)
library(stargazer)
library(flextable)
library(officer)
library(stringr)
library(pandoc)
library(webshot2)

setwd("C:/Users/kevin/Documents/QRM3")

# Load dataset & rename country variable bc it's confusing AF
pisa <- read_csv("QRMIII_Casedata_PISA.csv", show_col_types = FALSE) |>
  mutate(
    country_name = as.factor(CNTRYID)
  )

#Question 0
countries <- c("Mexico","Republic of Moldova","Croatia","Dominican Republic",
               "Türkiye","Indonesia","Japan","Montenegro")

pisa_sub <- pisa |>
  filter(country_name %in% countries) |>
```

```
  mutate(
    math_std    = as.numeric(scale(MATHSCORE)),
    reading_std = as.numeric(scale(READINGSCORE)),
    science_std = as.numeric(scale(SCIENCESCORE))
  )

#Sanity check to see whether there are indeed 8 countries
country_table <- pisa_sub |>
  count(country_name) |>
  arrange(country_name)
country_table

fig0 <- pisa_sub |>
  count(country_name) |>
  ggplot(aes(
    x = reorder(country_name, n),
    y = n,
    fill = country_name
  )) +
  geom_col() +
  geom_text(
    aes(label = n),
    hjust = 1.1,
    color = "white",
    size = 3.5
  ) +
  coord_flip() +
  scale_fill_manual(values = c(
    "steelblue","firebrick","seagreen","salmon2",
    "purple3","darkgoldenrod2","burlywood4","gray87"
  )) +
  labs(
    title = "Student composition by country",
    x = "Country",
    y = "Number of students",
    caption = paste0(
      "**Figure 0:** Student composition by country.<br>",
      "*Note:* The figure reports the number of students in the PISA 2022 sample for each
country included in the analysis."
    )
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.caption = ggtext::element_textbox_simple(
      size = 9,
```

```
      hjust = 0,
      width = unit(1, "npc"),
      margin = margin(t = 10)
    ),
    plot.margin = margin(t = 10, r = 10, b = 35, l = 10)
  )

fig0

ggsave(
  "figure0_student_composition.png",
  plot = fig0,
  width = 7,
  height = 5.5,
  dpi = 300
)

#Question 1
#Standardize the three test scores (z-scores)
# function: scale() standardizes using (x - mean(x)) / sd(x) and handles NA values.
pisa_sub <- pisa_sub |>
  mutate(
    math_std    = as.numeric(scale(MATHSCORE)),
    reading_std = as.numeric(scale(READINGSCORE)),
    science_std = as.numeric(scale(SCIENCESCORE))
  )

#Sanity check for the standardized scores:
sanity_check_stdscores <- tibble::tibble(
  Test_score = c("Mathematics", "Reading", "Science"),
  Mean_standardized = c(
    mean(pisa_sub$math_std, na.rm = TRUE),
    mean(pisa_sub$reading_std, na.rm = TRUE),
    mean(pisa_sub$science_std, na.rm = TRUE)
  ),
  SD_standardized = c(
    sd(pisa_sub$math_std, na.rm = TRUE),
    sd(pisa_sub$reading_std, na.rm = TRUE),
    sd(pisa_sub$science_std, na.rm = TRUE)
  )
)

sanity_check_stdscores

# 2) Summary statistics (mean + SD), original and standardized
q0_stats <- tibble::tibble(
```

```
  Score = c("Math", "Reading", "Science"),
  Mean_original = round(c(
    mean(pisa_sub$MATHSCORE, na.rm = TRUE),
    mean(pisa_sub$READINGSCORE, na.rm = TRUE),
    mean(pisa_sub$SCIENCESCORE, na.rm = TRUE)
  ),2),
  SD_original = round(c(
    sd(pisa_sub$MATHSCORE, na.rm = TRUE),
    sd(pisa_sub$READINGSCORE, na.rm = TRUE),
    sd(pisa_sub$SCIENCESCORE, na.rm = TRUE)
  ),2),
  Mean_standardized = round(c(
    mean(pisa_sub$math_std, na.rm = TRUE),
    mean(pisa_sub$reading_std, na.rm = TRUE),
    mean(pisa_sub$science_std, na.rm = TRUE)
  ),2),
  SD_standardized = round(c(
    sd(pisa_sub$math_std, na.rm = TRUE),
    sd(pisa_sub$reading_std, na.rm = TRUE),
    sd(pisa_sub$science_std, na.rm = TRUE)
  ),2)
)

ft_q0 <- flextable(q0_stats) |>
  autofit() |>
  add_footer_lines(
    values = "Note: Standardized scores are calculated using Z = (X − mean(X)) / sd(X).
After standardization the mean equals 0 and the standard deviation equals 1."
  )

save_as_docx(
  "Table 1: Summary Statistics of Original and Standardized Test Scores" = ft_q0,
  path = "C:/Users/kevin/Documents/QRM3/q0_summary_table.docx"
)

pisa_sub |>
  pivot_longer(
    cols = c(math_std, reading_std, science_std),
    names_to = "subject",
    values_to = "zscore"
  ) |>
  ggplot(aes(x = zscore, fill = subject)) +
  geom_histogram(
    bins = 30,
    color = "black",
    linewidth = 0.25
```

```
  ) +
  facet_wrap(~ subject, scales = "free_y") +
  scale_fill_manual(
    values = c(
      "math_std" = "steelblue",
      "reading_std" = "firebrick",
      "science_std" = "seagreen"
    )
  ) +
  labs(
    title = "Distribution of Standardized Test Scores",
    x = "Standard Deviations from the Sample Mean",
    y = "Count"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

note_text_fig1 <- paste0(
  "**Figure 1.** ",
  "<br>",
  "Note: The figure shows the distribution of standardized mathematics, reading, and
science scores.",
  "<br>",
  "Standardization is performed using z-scores, resulting in a mean of zero and a standard
deviation of one."
)

fig1_q1 <- pisa_sub |>
  pivot_longer(
    cols = c(math_std, reading_std, science_std),
    names_to = "subject",
    values_to = "zscore"
  ) |>
  ggplot(aes(x = zscore, fill = subject)) +
  geom_histogram(
    bins = 30,
    color = "black",
    linewidth = 0.25
  ) +
  facet_wrap(~ subject, scales = "free_y") +
  scale_fill_manual(
    values = c(
      "math_std" = "steelblue",
      "reading_std" = "firebrick",
      "science_std" = "seagreen"
    )
```

```
  ) +
  labs(
    title = "Distribution of Standardized Test Scores",
    x = "Standard Deviations from the Sample Mean",
    y = "Count",
    caption = note_text_fig1
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    plot.caption = ggtext::element_textbox_simple(
      size = 9,
      hjust = 0,
      halign = 0,
      width = unit(1, "npc"),
      padding = margin(0, 0, 0, 0),
      margin = margin(t = 10)
    ),
    plot.margin = margin(t = 10, r = 10, b = 35, l = 10)
  )

ggsave(
  "figure1_scores.png",
  plot = fig1_q1,
  width = 8,
  height = 6.8,
  dpi = 300
)

#Question 2
pisa_sub |>
  ggplot(aes(x = reading_std, y = math_std)) +
  geom_point(alpha = 0.25, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", linewidth = 1.2) +
  labs(
    title = "Correlation between Standardized Reading and Mathematics Scores",
    x = "Standardized Reading Score",
    y = "Standardized Mathematics Score"
  ) +
  theme_minimal()

note_text_fig2 <- paste0(
  "**Figure 2.** ",
  "<br>",
  "Note: Each point represents a student in the sample. ",
```

```
  "The solid line shows the fitted linear relationship between standardized reading and
mathematics scores.",
  "Both variables are expressed as z-scores, meaning that values represent deviations from
the sample mean measured in standard deviations."
)

fig2_q2 <- pisa_sub |>
  ggplot(aes(x = reading_std, y = math_std)) +
  geom_point(alpha = 0.25, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "firebrick", linewidth = 1.2) +
  labs(
    title = "Correlation Between Standardized Reading and Mathematics Scores",
    x = "Standardized Reading Score",
    y = "Standardized Mathematics Score",
    caption = note_text_fig2
  ) +
  theme_minimal() +
  theme(
    plot.caption = ggtext::element_textbox_simple(
      size = 9,
      hjust = 0,
      halign = 0,
      width = unit(1, "npc"),
      padding = margin(0, 0, 0, 0),
      margin = margin(t = 10)
    ),
    plot.margin = margin(t = 10, r = 10, b = 40, l = 10)
  )

ggsave(
  "figure2_reading_math_correlation.png",
  plot = fig2_q2,
  width = 8,
  height = 6.8,
  dpi = 300
)

# Question 3
# OLS regression model
model_q3 <- lm(math_std ~ reading_std, data = pisa_sub)

# Q3A: Conventional OLS SE (homoskedasticity-assumption)
summary(model_q3)

# Q3B: Heteroskedasticity robustness test for SE (homoskedasticity-check)
coeftest(model_q3, vcov. = vcovHC(model_q3, type = "HC1"))
```

```
# Comparison for SE and Robust SE
table_q3_compare <- modelsummary(
  list(
    "Conventional SE" = model_q3,
    "Robust SE (HC1)" = model_q3
  ),
  vcov = list(
    NULL,                          # default OLS SE
    vcovHC(model_q3, type = "HC1")     # robust HC1 SE
  ),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "reading_std" = "Standardized Reading Score"
  ),
  gof_map = data.frame(
    raw = c("nobs", "r.squared", "adj.r.squared", "f.statistic"),
    clean = c("Observations", "R-squared", "Adjusted R-squared", "F-statistic"),
    fmt = c(0, 3, 3, 1)
  ),
  stars = TRUE,
  fmt = 3,
  title = "Table 2. Regression Results for Standardized Mathematics Scores",
  notes = c(
    "Notes:",
    "Model (1) reports conventional OLS standard errors.",
    "Model (2) reports heteroskedasticity-robust standard errors (HC1).",
    "The dependent variable is the standardized mathematics score."
  ),
  output = "flextable"
)

table_q3_compare <- table_q3_compare |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

table_q3_compare

# Export to Word
save_as_docx(
```

```
  "Table 2. Regression Results for Standardized Mathematics Scores" =
table_q3_compare,
  path = "C:/Users/kevin/Documents/QRM3/table2_q3_comparison.docx"
)

#Question 4
#Step 1: Check the language variable before using it
table(pisa_sub$ST022Q01TA, useNA = "ifany")

#Variable has three meaningful labels that we now need to convert into binary labels
pisa_q4 <- pisa_sub |>
  mutate(
    speaks_test_lang = case_when(
      ST022Q01TA == "Language of the test" ~ 1,
      ST022Q01TA == "Other language" ~ 0,
      TRUE ~ NA_real_
    )
  ) |>
  filter(!is.na(speaks_test_lang)) #If there are any missing values, exclude those
observations.

#sanity check
table(pisa_q4$speaks_test_lang)

model_q4 <- lm(
  math_std ~ reading_std + speaks_test_lang,
  data = pisa_q4
)
coeftest(  model_q4,  vcov. = vcovHC(model_q4, type = "HC1"))

table_q4 <- modelsummary(
  model_q4,
  vcov = vcovHC(model_q4, type = "HC1"),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "reading_std" = "Standardized Reading Score",
    "speaks_test_lang" = "Speaks Test Language at Home"
  ),
  gof_map = data.frame(
    raw = c("nobs", "r.squared", "adj.r.squared"),
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt = c(0, 3, 3)
  ),
  stars = TRUE,
  fmt = 3,
  title = "Table 3. Multivariate Regression Results for Standardized Mathematics Scores",
```

```
  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the standardized mathematics score.",
    "The variable 'Speaks Test Language at Home' equals 1 if the student speaks the test
language at home and 0 otherwise."
  ),
  output = "flextable"
)

table_q4 <- table_q4 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

table_q4

save_as_docx(
  "Table 3. Multivariate Regression Results for Standardized Mathematics Scores" =
table_q4,
  path = "C:/Users/kevin/Documents/QRM3/table3_q4_multivariate_regression.docx"
)

#Question 5
#Does the relationship between reading and math depend on whether the student speaks
the test language at home?
model_q5 <- lm(
  math_std ~ reading_std * speaks_test_lang,
  data = pisa_q4
)

coeftest(  model_q5,  vcov. = vcovHC(model_q5, type = "HC1"))
table_q5 <- modelsummary(
  model_q5,
  vcov = vcovHC(model_q5, type = "HC1"),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "reading_std" = "Standardized Reading Score",
    "speaks_test_lang" = "Speaks Test Language at Home",
    "reading_std:speaks_test_lang" = "Reading × Speaks Test Language"
  ),
  gof_map = data.frame(
```

```
    raw = c("nobs", "r.squared", "adj.r.squared"),
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt = c(0, 3, 3)
  ),
  stars = TRUE,
  fmt = 3,
  title = "Table 4. Interaction Regression Results for Standardized Mathematics Scores",
  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the standardized mathematics score.",
    "The variable 'Speaks Test Language at Home' equals 1 if the student speaks the test
language at home and 0 otherwise.",
    "The interaction term captures whether the relationship between reading and
mathematics differs by language spoken at home."
  ),
  output = "flextable"
)

table_q5 <- table_q5 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

# Preview in RStudio
table_q5

# Export to Word
save_as_docx(
  "Table 4. Interaction Regression Results for Standardized Mathematics Scores" =
table_q5,
  path = "C:/Users/kevin/Documents/QRM3/table4_q5_interaction_regression.docx"
)

note_text_fig3 <- str_wrap(
  "Note: Each point represents a student. The fitted lines show the estimated linear
relationship between standardized reading and mathematics scores for students who
speak the test language at home and those who do not.",
  width = 110
)

fig3_q5 <- pisa_q4 |>
```

```
  ggplot(aes(
    x = reading_std,
    y = math_std,
    color = factor(speaks_test_lang)
  )) +
  geom_point(alpha = 0.15, size = 0.8) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 1.2) +
  scale_color_manual(
    values = c("0" = "firebrick", "1" = "navy"),
    labels = c("0" = "Other language", "1" = "Test language")
  ) +
  labs(
    title = "Heterogeneous Correlation Between Reading and Mathematics Scores",
    subtitle = "Separate fitted lines by language spoken at home",
    x = "Standardized Reading Score",
    y = "Standardized Mathematics Score",
    color = "Language at home",
    caption = note_text_fig3
  ) +
  theme_minimal() +
  theme(
    plot.caption = element_text(hjust = 0, size = 9, margin = margin(t = 10)),
    plot.margin = margin(t = 10, r = 10, b = 25, l = 10)
  )

fig3_q5

ggsave(
  filename = "figure3_q5_interaction.png",
  plot = fig3_q5,
  width = 8,
  height = 6.5,
  dpi = 300
)

#Tables Part 1
model_q3 <- lm(math_std ~ reading_std, data = pisa_sub)

model_q4 <- lm(
  math_std ~ reading_std + speaks_test_lang,
  data = pisa_q4
)

model_q5 <- lm(
  math_std ~ reading_std * speaks_test_lang,
  data = pisa_q4
```

```
)

se_q3 <- sqrt(diag(vcovHC(model_q3, type = "HC1")))
se_q4 <- sqrt(diag(vcovHC(model_q4, type = "HC1")))
se_q5 <- sqrt(diag(vcovHC(model_q5, type = "HC1")))

stargazer(
  model_q3, model_q4, model_q5,
  type = "html",
  out = "C:/Users/kevin/Documents/QRM3/table_part1.html",
  se = list(se_q3, se_q4, se_q5),
  title = "Table 5. Regression Results for Standardized Mathematics Scores",
  dep.var.labels = "Standardized Mathematics Score",
  column.labels = c("Q3", "Q4", "Q5"),
  covariate.labels = c(
    "Standardized Reading Score",
    "Speaks Test Language at Home",
    "Reading × Speaks Test Language"
  ),
  omit.stat = c("f", "ser"),
  digits = 3,
  no.space = TRUE,
  notes = "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.
The variable 'Speaks Test Language at Home' equals 1 if the student speaks the test
language at home and 0 otherwise.",
  notes.align = "l"
)

webshot(
  url = "C:/Users/kevin/Documents/QRM3/table_part1.html",
  file = "C:/Users/kevin/Documents/QRM3/table5_part1.png",
  vwidth = 1400,
  vheight = 900,
  zoom = 2
)

# Question 6: ICT resources and standardized mathematics scores
# Check availability of ICTRES by country
table(!is.na(pisa_sub$ICTRES), pisa_sub$country_name)

# Keep complete observations for relevant variables
pisa_sub_complete1 <- pisa_sub |>
  drop_na(math_std, ICTRES, country_name)

# Check factor levels to identify the baseline country
levels(pisa_sub_complete1$country_name)
```

```
# Estimate regression model
std_MathScore_ICT_model <- lm(
  math_std ~ ICTRES + country_name,
  data = pisa_sub_complete1
)

summary(std_MathScore_ICT_model)

coeftest(
  std_MathScore_ICT_model,
  vcov. = vcovHC(std_MathScore_ICT_model, type = "HC1")
)

table_q6 <- modelsummary(
  std_MathScore_ICT_model,
  vcov = vcovHC(std_MathScore_ICT_model, type = "HC1"),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "ICTRES" = "ICT Resources at Home"
  ),
  coef_omit = "country_name",
  gof_map = data.frame(
    raw = c("nobs", "r.squared", "adj.r.squared"),
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt = c(0, 3, 3)
  ),
  stars = TRUE,
  fmt = 3,
  title = "Table 6. Regression Results for Standardized Mathematics Scores and ICT
Resources",
  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the standardized mathematics score.",
    "Country dummies are included in the regression but omitted from the table."
  ),
  output = "flextable"
)

table_q6 <- table_q6 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
```

```
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

table_q6

save_as_docx(
  "Table 6. Regression Results for Standardized Mathematics Scores and ICT Resources"
= table_q6,
  path = "C:/Users/kevin/Documents/QRM3/table6_q6_ict_math.docx"
)

# Question 7
# Add log-transformed math score and keep complete observations
pisa_sub_complete1 <- pisa_sub |>
  mutate(
    logged_MathScores = log(MATHSCORE)
  ) |>
  drop_na(logged_MathScores, ICTRES, country_name)

logged_MathScore_ICT_model <- lm(
  logged_MathScores ~ ICTRES + country_name,
  data = pisa_sub_complete1
)

summary(logged_MathScore_ICT_model)

coeftest(
  logged_MathScore_ICT_model,
  vcov. = vcovHC(logged_MathScore_ICT_model, type = "HC1")
)

table_q7 <- modelsummary(
  logged_MathScore_ICT_model,
  vcov = vcovHC(logged_MathScore_ICT_model, type = "HC1"),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "ICTRES" = "ICT Resources at Home"
  ),
  coef_omit = "country_name",
  gof_map = data.frame(
    raw = c("nobs", "r.squared", "adj.r.squared"),
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt = c(0, 3, 3)
  ),
  stars = TRUE,
  fmt = 3,
```

```
  title = "Table 7. Regression Results for Log Mathematics Scores and ICT Resources",
  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the log-transformed mathematics score.",
    "Country dummies are included in the regression but omitted from the table."
  ),
  output = "flextable"
)

table_q7 <- table_q7 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

table_q7

save_as_docx(
  "Table 7. Regression Results for Log Mathematics Scores and ICT Resources" =
table_q7,
  path = "C:/Users/kevin/Documents/QRM3/table7_q7_logged_math.docx"
)

# Question 8
# Check variable type and categories
class(pisa_sub$ST258Q01JA)
unique(pisa_sub$ST258Q01JA)

# Create factor and drop missing observations
pisa_sub_complete2 <- pisa_sub_complete1 |>
  mutate(
    food_shortage = as.factor(ST258Q01JA)
  ) |>
  drop_na(math_std, ICTRES, country_name, food_shortage)

# Check baseline country
levels(pisa_sub_complete2$country_name)

# Estimate model with poverty proxy
controlVar_MathScore_ICT_model <- lm(
  math_std ~ ICTRES + country_name + food_shortage,
  data = pisa_sub_complete2
```

```
)

summary(controlVar_MathScore_ICT_model)

coeftest(
  controlVar_MathScore_ICT_model,
  vcov. = vcovHC(controlVar_MathScore_ICT_model, type = "HC1")
)

table_q8 <- modelsummary(
  controlVar_MathScore_ICT_model,
  vcov = vcovHC(controlVar_MathScore_ICT_model, type = "HC1"),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "ICTRES" = "ICT Resources at Home"
  ),
  coef_omit = "country_name|food_shortage",
  gof_map = data.frame(
    raw = c("nobs", "r.squared", "adj.r.squared"),
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt = c(0, 3, 3)
  ),
  stars = TRUE,
  fmt = 3,
  title = "Table 8. Regression Results with Food Availability Control",
  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the standardized mathematics score.",
    "Country dummies and food availability categories are included in the regression.",
    "Country dummy coefficients are omitted from the table."
  ),
  output = "flextable"
)

table_q8 <- table_q8 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

table_q8
```

```
save_as_docx(
  "Table 8. Regression Results with Food Availability Control" = table_q8,
  path = "C:/Users/kevin/Documents/QRM3/table8_q8_control_food.docx"
)

# Question 9 and 10
# Create instrument and drop missing observations
pisa_sub_complete3 <- pisa_sub_complete2 |>
  mutate(
    parent_edu = as.factor(HISCED)
  ) |>
  drop_na(math_std, ICTRES, country_name, food_shortage, parent_edu)

# Check baseline country
levels(pisa_sub_complete3$country_name)

#Check dataset size and levels
dim(pisa_sub_complete3)
nlevels(as.factor(pisa_sub_complete3$country_name))
nlevels(as.factor(pisa_sub_complete3$food_shortage))
nlevels(as.factor(pisa_sub_complete3$parent_edu))

# Estimate IV model
instrumentVar_MathScore_ICT_model <- ivreg(
  math_std ~ ICTRES + country_name + food_shortage |
    parent_edu + country_name + food_shortage,
  data = pisa_sub_complete3
)

summary(instrumentVar_MathScore_ICT_model, diagnostics = FALSE)

coeftest(
  instrumentVar_MathScore_ICT_model,
  vcov. = vcovHC(instrumentVar_MathScore_ICT_model, type = "HC1")
)

table9_q10 <- modelsummary(
  instrumentVar_MathScore_ICT_model,
  vcov = vcovHC(instrumentVar_MathScore_ICT_model, type = "HC1"),
  coef_rename = c(
    "(Intercept)" = "Constant",
    "ICTRES" = "ICT Resources at Home"
  ),
  coef_omit = "country_name|food_shortage|parent_edu",
  gof_map = data.frame(
    raw = c("nobs", "r.squared", "adj.r.squared"),
```

```
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt = c(0, 3, 3)
  ),
  stars = TRUE,
  fmt = 3,
  title = "Table 9. Instrumental Variables Regression Results for Mathematics Scores",
  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the standardized mathematics score.",
    "ICT Resources at Home is instrumented using parental education.",
    "Country dummies, food availability categories, and parental education dummies are
included in the estimation but omitted from the table."
  ),
  output = "flextable"
)

table9_q10 <- table9_q10 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
  padding(padding = 4, part = "all")

table9_q10

# Export to Word
save_as_docx(
  "Table 9. Instrumental Variables Regression Results for Mathematics Scores" =
table_q10,
  path = "C:/Users/kevin/Documents/QRM3/table9_q10_iv_regression.docx"
)

se_q6  <- vcovHC(std_MathScore_ICT_model, type = "HC1")
se_q7  <- vcovHC(logged_MathScore_ICT_model, type = "HC1")
se_q8  <- vcovHC(controlVar_MathScore_ICT_model, type = "HC1")
se_q10 <- vcovHC(instrumentVar_MathScore_ICT_model, type = "HC1")

table_part2 <- modelsummary(
  list(
    "Q6"  = std_MathScore_ICT_model,
    "Q7"  = logged_MathScore_ICT_model,
    "Q8"  = controlVar_MathScore_ICT_model,
    "Q10" = instrumentVar_MathScore_ICT_model
```

```
  ),

  vcov = list(
    se_q6,
    se_q7,
    se_q8,
    se_q10
  ),

  coef_rename = c(
    "(Intercept)" = "Constant",
    "ICTRES" = "ICT Resources at Home"
  ),

  coef_omit = "country_name|parent_edu",

  gof_map = data.frame(
    raw   = c("nobs", "r.squared", "adj.r.squared"),
    clean = c("Observations", "R-squared", "Adjusted R-squared"),
    fmt   = c(0, 3, 3)
  ),

  stars = TRUE,
  fmt = 3,

  title = "Table Part 2: Regression Results",

  notes = c(
    "Notes:",
    "Heteroskedasticity-robust standard errors (HC1) are reported in parentheses.",
    "The dependent variable is the standardized mathematics score in Q6, Q8, and Q10.",
    "In Q7 the dependent variable is the log-transformed mathematics score.",
    "Country dummies are included in all specifications.",
    "Food shortage and parental education controls are included where relevant."
  ),

  output = "flextable"
)

table_part2 <- table_part2 |>
  autofit() |>
  theme_booktabs() |>
  align(align = "center", part = "all") |>
  align(j = 1, align = "left", part = "body") |>
  bold(part = "header") |>
  fontsize(size = 10, part = "all") |>
```

```
  padding(padding = 4, part = "all")

table_part2

save_as_docx(
  "Table Part 2: Regression Results" = table_part2,
  path = "C:/Users/kevin/Documents/QRM3/table_part2_regressions.docx"
)
```